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REVIEW

The Versatility of the Taguchi Method: Optimizing Experiments Across Diverse Disciplines

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Abstract

The Taguchi method, a robust experimental design technique, establishes a strong connection between input and output variables. Known for its capacity to yield precise results with fewer trials and minimized errors, this method has gained widespread application in various fields such as engineering, physics, chemistry, economics, finance, and more. In this paper, the authors examine the importance of the Taguchi orthogonal array method, its step-by-step optimization procedure, and its potential for future applications. Through a thorough literature review, the authors investigate how the Taguchi method has been effectively employed to identify key factors influencing response variables. The versatility of the Taguchi method becomes apparent when considering its applications across diverse disciplines. Researchers in engineering have successfully utilized this technique to optimize processes and enhance product quality. Furthermore, in scientific fields like physics and chemistry, the Taguchi method has proven invaluable for conducting experiments efficiently, resulting in more accurate and reproducible outcomes. Researchers gain critical insights into the effects of factors on the response variable by employing statistical tools such as mean analysis, variance analysis, and signal-to-noise ratio. The Taguchi method remains a valuable and broadly applicable tool for optimizing experiments and identifying influential factors across multiple disciplines. This paper's extensive literature review emphasizes its significance in various fields and outlines the step-by-step procedure to leverage its potential for optimization.

Keywords Taguchi method · Analysis of mean (ANOM) · Analysis of variance (ANOVA) · Signal-to-noise ratio

Abbreviations

ANOVA	Analysis of variance
ANOM	Analysis of mean

Extended author information available on the last page of the article

S/N Ratio	Signal to noise ratio
DOE	Design of experiment
FFD	Full factorial design
OA	Orthogonal array
EO	Energy optimization
MWNT	epoxy/nanoclay/multi-walled carbon nanotube
V/V	Anti-solvent to solvent
UPV	Ultrasonic pulse velocity

1 Introduction

Almost every research project begins with an experiment aimed at exploring something new of interest. The primary motive behind an experiment is to draw inferences about the population being studied. During a statistical investigation, a researcher seeks to achieve sound and practically relevant results. To uncover the hidden characteristics of a population (such as the influence of various factors on a specific response variable), it is essential to create an experimental setup that allows for a more accurate exploration of these characteristics. In statistical literature, there exists a field known as the design of experiments (DOE), which focuses on studying different types of experimental setups, including full factorial design (FFD), fractional design, Latin square method, Taguchi design method, and others. The goal is to minimize error in predicting the response and to draw valid conclusions regarding the impact of different factors on an underlying response variable, commonly referred to as yield/output.

DOE is a widely utilized statistical technique among researchers across various fields to identify the impact of different factors on the output process. The application of statistics in general, and DOE in particular, is so extensive that there is hardly any field where it does not find its use. It encompasses a broad range of applications and has been rigorously tested by numerous scientists and researchers, including those in agriculture [1], social science [2], chemistry [3], biology [4], service sectors [5], manufacturing [6], and economics [7], among others. This technique elucidates the relationship between variables, specifically input and output variables. It is employed to enhance systems (which may be simple or complex), develop new products, and improve existing products and processes. The DOE technique facilitates economical experimentation, from information collection to identifying primary objectives and drawing conclusions from the investigation. Its purpose is to gather data that can be analyzed using statistical tools for maximum or optimal output.

DOE methodology was initially introduced by Sir Ronald Fisher in his groundbreaking works, "The Arrangement of Field Experiments" (1926) and "The Design of Experiments" (1935) [8, 9]. His significant contribution enabled researchers to manipulate multiple factors simultaneously and introduced ANOVA, a statistical tool that assesses whether there are any significant differences among groups [10, 11]. The study analyzed agricultural output with the aim of examining the simultaneous effects of various factors on it. He assessed the proportions of light, fertilizers, water, and other inputs necessary to achieve exceptional agricultural results. Since the introduction of DOE, extensive research has been conducted on both its application and advancement.

In the 19th century, a DOE known as factorial designs was utilized by Joseph Henry Gilbert and Bennet Lawes of the Rothamsted Experimental Station [12]. R. A. Fisher contended in 1926 that "complex" designs such as FFD were less significant than examining one factor at a time. An FFD identifies all possible combinations of levels across all factors and experiments with these combinations. Such experiments demand a considerable number of trials, resources, and time to execute. When conducting a large number of experiments, the results tend to become inaccurate and heterogeneous. For instance, in a study with r factors and n levels, one would require n^r trials. With large values of n and r, the number of trials becomes relatively substantial. When the number of factors and their levels increases, a researcher may find the FFD unaffordable due to the higher number of replications involved.

To address this issue, statisticians develop a design known as fractional design, which is carefully selected from FFD based on the "sparsity-of-effects" principle. The sparsity-of-effects guideline indicates that the main effects and lower-order interactions typically dominate the system's arrangement. A fractional design consists of a carefully chosen fraction of the experimental trials, where some combinations (generally at least half) are omitted [13]. For instance, in the mentioned study, if r factors are varied at n levels, the fractional design requires $n^{(r-1)}$ experiments, resulting in the number of trials being less than half of FFD. Fractional design necessitates a solid statistical understanding for experimentation, and there are no established rules for its analysis to draw conclusions [13, 14]. When resources are limited (or when the number of factors and their levels is large), it is advisable to use a fractional design for the experiment, as only a selected subset of runs is utilized. Typically, researchers conduct a subset of experiments from FFD to conserve time and resources. Consequently, selecting specific experiments from the experimental setup is challenging for nearly every researcher. It requires proper knowledge to extract the subset of experiments from FFD.

There exists another method known as the Taguchi method (introduced by G. Taguchi in the 1940s), which is one of the most effective techniques for reducing the number of trials based on orthogonal arrays (OAs). This method offers researchers significantly reduced variance for experiments with optimal settings of control input parameters [15]. Taguchi employed partial factorial design to minimize the number of trials, which consisted of OAs that provide the minimum number of trials for a given set of factors and levels [15]. The influence of various factors on performance characteristics can be assessed through the orthogonal array (OA) experimental design using a condensed set of trials. Taguchi (1940) established a unique set of general design guidelines for a factorial design method applicable to numerous scenarios, known as the Taguchi method, aimed at minimizing the number of experiments (i.e., reducing the quantum of resources – material, manpower, equipment, and time). Athreya and Venkatesh (2012) utilized both FFD and the Taguchi method for their experimental setup, finding that both yielded the same results [16].

Taguchi method aids in enhancing the quality of existing products/processes as well as in developing new ones. Taguchi established a collection of partial factorial experimental matrices, referred to as OAs, which are utilized in various situations.

An OA is a set of well-balanced (minimum) trials. The Taguchi method is traditionally a sensitivity analysis technique, where a selected factor is varied at different levels while keeping other factors constant. In Taguchi methodology, all components are adjusted simultaneously according to predefined tables known as “OAs.” Choosing the appropriate OA for a specific problem is termed “experiment design.”

The selection of the Taguchi OA matrix is based on the input factors and levels. A key aspect of the Taguchi method is the selection of appropriate OAs. Taguchi simplifies the assignment of factors through the use of triangular tables and linear graphs [17]. Researchers can utilize the Taguchi method without requiring advanced statistical knowledge. It is a powerful technique that is comprehensive and easy to apply, offering an efficient, systematic, and straightforward approach to optimizing designs for quality, cost, and performance [18]. Analyzing the results using the Taguchi method provides sufficient and more accurate information with a minimal number of experiments. This method is employed to assess the primary impacts of factors on the output variable.

DOE provides the information that is required to manage the input process to optimize the output variable. It will give the most sensible information concerning the input factors of the output variable. Conducting a large number of experiments, the result tends to become inaccurate and heterogeneous [19]. As mentioned earlier the Taguchi method DOE can be used in every field to measure the relationship between the input and the output variables such as health care [20], Biotechnology [21], advertising and marketing [22], engineering [23] and so on. One of the best DOE with a minimum number of trials is the Taguchi method. Table 1 gives the difference between the FFD and the Taguchi method.

1.1 Objectives

This study explores the effects of different input factors on response variables. The current research has been carried out to achieve the following objectives.

1. To provide a comprehensive, step-by-step guide for implementing the Taguchi method in quality improvement processes, ensuring practitioners can effectively enhance product or process quality.

Table 1 Taguchi method versus full factorial design

Number of factors	Number of levels	Number of runs as per Taguchi OA	Number of runs as per FFD
3	2	4	8
7	2	8	128
11	2	12	2048
15	2	16	32,768
4	3	9	81
5	4	16	1024

Look at Table 1 and see the difference between the number of experiments per the Taguchi method and the FFD. The Taguchi method reduces our time and resources and it provides an approximate result

2. To identify and explore the various disciplines and scenarios where the Taguchi method can be applied, offering valuable insights for researchers and industry professionals seeking to optimize their processes.

2 Taguchi Method

Dr. G. Taguchi of the Nippon Telegraph and Telephone Industry in Japan has developed a DOE based on orthogonal arrays, which significantly reduces variance for experiments with optimal settings of control input parameters. It also provides a set of balanced (minimum) trials and a ratio known as signal-to-noise, which is the logarithmic function of the output value that aids in data analysis and predicts optimal results.

The Taguchi method addresses optimization settings in two categories: static problems and dynamic problems. Generally, the static problem procedure involves different control factors that directly select the desired or target output value. The optimization process then entails determining the best control factor levels to achieve the desired output. This type of issue is referred to as a “static problem.” In contrast, the dynamic problem procedure involves a signal input that directly influences the result, and the optimization process focuses on identifying the best control factor levels so that the “input signal/result” ratio is closest to the desired relationship. This process is termed a “dynamic problem.”

The selection of the Taguchi orthogonal array is based on the input factors and their levels. A key aspect of the Taguchi method is selecting suitable orthogonal arrays and interactions for the appropriate columns. Taguchi simplifies the assignment of factors through the use of triangular tables and linear graphs. ANOVA is employed to measure the percentage contribution of each factor to the response variable across levels, helping to identify which factors require more or less control [24]. The response means or ANOM can also be utilized to determine which factor has the greatest impact on the response variable. The signal-to-noise (S/N) ratio is used to assess the deviation of performance characteristics. Generally, there are three types of performance characterises in the analysis of S/N ratio [25].

- Smaller the better.

$$\frac{S}{N} = -10\log\left(\frac{1}{n}*\sum{}_{k=1}^{n}y_k^2\right) \tag{2.1}$$

- Larger the better.

$$\frac{S}{N} = -10\log\left(\frac{1}{n}*\sum{}_{k=1}^{n}1/y_k^2\right) \tag{2.2}$$

- Nominal the better.

$$\frac{S}{N} = -10\log\left(\frac{\bar{Y}^2}{s_k^2}\right) \tag{2.3}$$

Where:
y_k : Output values
n : Repeated number of trails
$\bar{Y}$: Mean of output values
s_k^2 : Variance

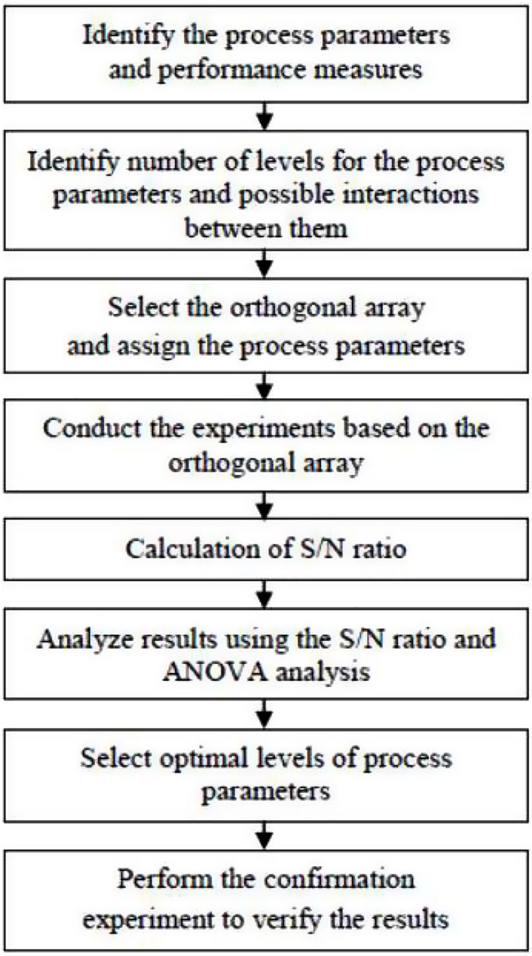


Fig. 1 Steps in the Taguchi method

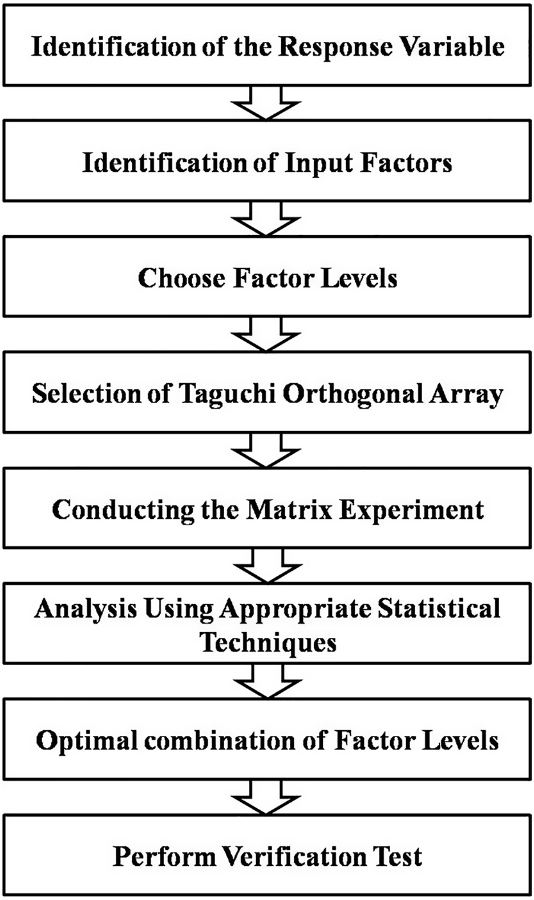
For each type of characteristic, with the above S/N transformation, the higher the S/N ratio the better the result. Optimization of performance measures using the parameter design of the Taguchi method is summarized in the steps as shown in Fig. 1 [26].

3 Methodology

The Taguchi method depends on the following steps:

I. Select the control input parameters, the response variable and levels: It is a researcher's first aim to define the problem and choose the parameters and the definite number of levels.
II. Choose an orthogonal Array: Selecting the orthogonal array table is based on the input parameters and their levels as shown in Fig. 2.
III. Assignment of the parameters: One of the important works of the Taguchi method is to select suitable orthogonal arrays and interactions to the appropriate vertical

Fig. 2 Array selector



columns. Taguchi makes an assignment of the factors simple by the use of Triangular tables and linear graphs.

IV. Experiment: Once the orthogonal array table is selected, the investigations are led according to the level combination. Fundamentally, every one of the analyses is directed. The connection segments and dummy variable segments will not be considered for experimenting (sometimes researchers consider both dummies as well as interaction). The exhibition parameter under investigation is noted down for each examination to direct the affectability examination.

V. Analy. sis: After experimenting, it is necessary to analyse the output values. Different statistical tools can be used for analysis such as ANOVA, S/N ratio, ANOM, regression etc.

VI. Confirmation: Finally, we have to check whether our optimal combination of parameters at different levels is accurate or not.

3.1 Overview of Taguchi's Method of Experiment

The experiment begins by selecting various factors and their levels that influence a specific response variable. Given the different types of Taguchi orthogonal arrays (OAs), it is essential to choose an appropriate array for experimentation. The selection of an OA is determined by the number of levels and factors involved. The goal of employing the Taguchi method is to identify which factor has the greatest impact on the response variable and to find the optimal combination of parameters across various levels. Figure 3 illustrates the procedure followed in the Taguchi design. It has been crafted to offer guidelines and direction for successfully achieving the best possible results.

3.1.1 Selection of Factors, Response Variables and Levels

The first step is to determine the response variables. The result that will be observed from the experiment is called the response. These factors are called the input factors. The next step is to choose the number of levels that is the number of datasets selected for experimentation. The number of factors and their levels is decided based on resource availability and time for experimentation. It is worth mentioning that there

Number of Levels	Maximum Number of Factors														
	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
2	L4	L4	L8	L8	L8	L8	L12	L12	L12	L12	L16	L16	L16	L16	L32
3	L9	L9	L9	L18	L18	L18	L18	L27	L27	L27	L27	L27	L36	L36	L36
4	L16	L16	L16	L16	L32	L32	L32	L32	L32	–	–				
5	L25	L25	L25	L25	L25	L50	L50	L50	L50	L50	L50				

Fig. 3 Flow diagram of Taguchi approach

should minimum of two levels for estimating the factor affecting the response variable because no algorithm can extract any information from a single-level dataset. A minimum of two levels should be used if it is known that there exists a direct linear relationship between input factors and the response variable to estimate the factor effect on the response variable. Apart from these, any level of data set is selected like three, four, five, and so on.

3.1.2 Selection of Suitable Experimental Design

Taguchi gives numerous standard OAs. An OA is a set of well-balanced (minimum) trials. It is a subset of the factors space that represents a large number of decision variables. Figure 2 shows how to select OA, it depends on the maximum number of factors and the number of its levels. if four factors are varied at three levels and it shows that that OA L9, L18, and L27 are enough to experiment [27–29].

The selection of a suitable OA depends on the underlined input factors and their levels.

The OA is a table of columns and rows (matrix) and a subset of combinations of various input factors at different levels. The only aim of OA is to measure the impacts of different input factors on the response process with a fewer number of runs. The fewer number of experiments as per the Taguchi method is calculated as:

$$N_t = 1 + N_f\,(N_l - 1) \tag{3.1}$$

Where N_t, N_f and N_l respectively represent the required number of experiments, the total number of factors, and the total number of levels. For example, if we have $N_f = 4$, $N_l = 3$

$$N_t = 1 + 4\,(3 - 1) = 9$$

Therefore, for four factors at three levels, the minimum number of experiments required is nine as shown in Fig. 4. The following standard OAs are commonly used for DOE.

3.1.3 Conducting an Experiment

Once the OA is selected, the experiments are run as per the factor levels combination. Taguchi method can be used in any event where there is a controllable procedure. The controllable procedure can be computer models, real equipment tests, and a mathematical equation that can enough display the reaction of the procedure. Essentially substitutes the factors to the test setup structure and computes the experimental value.

Fig. 4 Standard orthogonal array's

2-level arrays	L4, L8, L12, L16, L32
3-level arrays	L9, L18, L27
4-level arrays	L16, L32

The test sheets show just the fundamental factor levels required for each trial. It is imperative to see that everything is being completed according to the arrangement. At the point when the issue is to think about the fundamental factors, the factors can be appointed in any request to every section of the OA. The Taguchi OA design obtains the output from each trail. The output obtained from every one of the trials of a trial is named as information or result.

3.1.4 Analysis and Prediction

After experimentation, the optimal combination among the underlined factor levels leading to the optimal value of the response is investigated by using appropriate statistical tools like ANOM and the percentage contribution due to different factors in the variation of response is analysed by ANOVA. Regression coefficients and ANOM are widely used to identify which factor impacts more on the response variable. While there are various possible average proportions, three of them are treated as standard and are commonly applied in the situations below:

I. Largest-is-best.
II. Smallest-is-best.
III. Nominal-is-best.

3.1.5 Confirmation Test

After the analysis is done using appropriate statistical tools the final step is experimental confirmation. The confirmation test is very important when the Taguchi method is considered because only the minimum number of trials is run. The main objective of this is to investigate the validity of the result derived from the experiment. It helps in proving whether the predicted optimal combination is accurate or not.

4 Literature Review

The Taguchi method establishes the relationship between input and output variables. It is utilized to enhance existing products or processes and to develop new ones. The DOE, along with appropriate statistical tools, compares results to identify the optimal treatment combination. Taguchi's method is grounded in OAs. The concept of the OA was introduced as an experimental design by C. R. Rao in 1946 [29]. To determine the optimal combination among factors with a limited number of factor combinations, OAs have gained widespread application in industrial experimentation. Taguchi conceived this idea during his visit to an Indian statistical institute and employed OAs to identify the best optimal combination of factors that yields high results and is resilient to environmental changes. As a mathematical invention, OAs were first reported in 1897 by the French mathematician Jacques Hadamard. Hadamard matrices are akin to Taguchi matrices, differing only in the arrangement of rows and columns. This plays a crucial role in the development of the Taguchi method. Taguchi utilized OAs to create a new factorial design that encompasses various applications. It was successfully adopted in Indian and Japanese industries and was later embraced by the US industry. One advantage of the Taguchi method is that it enables researchers and engineers to identify which factors most significantly affect the response variable, even with limited statistical knowledge and resources [30]. When engineers are tasked with implementing the Taguchi method in industry, it necessitates statistical analysis, teamwork, planning, communication, and engineering skills [31]. Taguchi seeks to achieve robust processes or products across various factors and levels while minimizing the number of experiments (short-term and low resource costs), unlike full factorial designs where every possible combination must be examined [32].

The Taguchi method was also applied to advertising and marketing [22], biotechnology [21], engineering [23], health care [20], improving forecasting rate [33], mining industry [34] so forth and it is used in various field such as the following.

Sun et al. (2018) investigated the effects of flow discharge, slope, and freeze-thaw cycles on soil detachment capacity. The Taguchi method was employed for the experimental setup to analyze the results using ANOVA and the signal-to-noise (S/N) ratio. It was determined that the factors of slope and flow discharge significantly influence soil detachment capacity [35].

Cabrars Rios et al. (2002) developed a new technique to finalize the design of cell manufacturing. The regression model and the Taguchi method were utilized to create this new approach. It was concluded that the Taguchi method is a singular and effective means to produce a cell design with a high profit margin and low sensitivity to noise variables [36].

Parinam et al. (2019) applied the Taguchi L9 design method to optimize parameters for high-transmission optical filters. The Taguchi L9 method was combined with a genetic algorithm to derive the optimal values of design parameters. The three parameters studied were material refractive index, thickness, and the number of levels. It was found that the thickness of the material significantly affects the response variable [37].

Ghosh and Mondal (2019) optimized the process parameters for a higher percentage of fluoride elimination from aqueous solutions using AI-IEBA in a system. The parameters affecting the response include adsorbent dose, initial concentration, pH, temperature, and contact time. It was found that contact time significantly impacts the response variable. As contact time increases, the percentage of removal gradually rises. The Taguchi method and S/N ratio were employed for the experimental setup and data analysis, respectively [38].

Mazumdar and Hoa (1995) investigated the effects of laser power, tap speed, and consolidation pressure on body strength. To determine which factors most influence the interplay bond strength, the Taguchi method was utilized for the minimum trials, and ANOVA was applied for data analysis. It was found that laser power significantly affects the interplay bond strength. The percentage contributions of the consolidation process, laser power, and tap speed to the interplay bond strength were 2.72%, 79.08%, and 17.05%, respectively [39].

Alharthi and Yang (2014) employed the Taguchi L9 method, regression, and ANOVA to examine which factors most influence extinguishing time and percentage damage (damage caused by fire). The factors affecting damage include the training of firemen, their experience, response to an alarm, age, and qualifications. It was found that the training and experience of firemen significantly impact extinguishing time and percentage damage [40].

Park et al. (2017) investigated the impact on powder binder injection moulding due to different parameters. The factors that impact powder binder separation are initial powder volume fraction, powder size, shear rate and injection temperature. The Taguchi method and S/N ratio were used to find which parameter impacts more on the response variable. It was found that the most important factor for powder binder separation is the initial powder volume [41].

Celik and Karatepe (2007) utilized the Taguchi strategy to assess and forecast bank crises by executing the neural networks. The artificial neural network which works with the banking information belonging to identical data and other artificial neural network displays that work with cross-sectional banking information has been drafted and tested. The ideal characteristics of these models have been directed by utilizing the Taguchi technique as a DOE [42].

Ketkar and Vaidya (2014) utilized a DOE known as the Taguchi method to determine the criteria for selecting MBA students. The parameters that affect the selection of MBA students are: marks 10th, marks 12th, work experience, entrance test score, scores at qualifying degree, personal interview score and group exercise score. It was found that the group exercise score, entrance test mark and personal interview score have positive effects on the selection of a candidate, whereas grades have a marginally positive effect and 10th, 12th and experience have negative gain. Hence the group exercise score, entrance test mark and interview score are studied to be effective for MBA candidate's selection. The universities and the institutes can choose better quality candidates with the help of this [43].

Madhav et al. (2017) identified the impacts of feed, material, cutting speed and depth of cut on hardness and surface roughness. The Taguchi method was used for DOE to find the optimal combination of factors the optimal value of hardness and surface roughness respectively [44].

Wang and Huang (2007) utilized the Taguchi method and ANOM to investigate which parameter is powerful in forecasting. The parameters are horizon length, data period and the number of observations required. The Taguchi method was used for experimental purposes and the ANOM was used to find which parameters impact more on the response variable. It was found that the horizon length is more significant than the data period and the number of observations in the preceding analysis–forecasting model [45].

Kim et al. (2019) investigated the impact of different parameters (revolutions per minute, the water content in the material and the diameter of the nozzle tip) to improve the 3D printability. It was found that the revolutions per minute affect more on the 3D printability while the diameter of the nozzle tip impacts less the improvement of 3D printability with the help of the Taguchi method [46].

Lin et al. (2009) utilized the Taguchi L8 OA and ANOVA to investigate the effect of the magnetic force of electrical discharge machining by various parameters. The parameters that affect the response variable are peak current, high voltage auxiliary current, machining polarity, pulse duration, servo reference voltage and non-load voltage. It was observed that the peak current and machining polarity were the significant parameters affecting magnetic force-assisted electrical discharge machining [47].

Lee and Teng (2009) used the Taguchi technique for budgetary data of Taiwan's electronic division to build up an early cautioning model with a highly significant discarnate capability, to give recommendations to financial specialists and loan bosses to secure their interest rates. They found three important factors, viz., return on asset, return on equity and debt to equity ratio in the financial crises early warning model [48].

Pander et al. (2018) implemented the Taguchi method to gauge the optical properties of a carbon nanotube forest by reducing the total ultraviolet-visible. The response variable was impacted by five parameters, viz., deposition bias, acetylene flow, catalyst thickness, acetylene to hydrogen gas flow ratio and a buffer layer. It was found that the most influential response variable in controlling the total ultraviolet-visible was deposition bias and the catalyst thickness [49].

Sangkharat and Dechjaren (2017) utilized the Taguchi method to investigate the impact of various parameters on the spinning process, with a particular emphasis on thickness variation and wrinkle severity. The study examined factors such as roller radius, spindle speed, depth, support roller width, friction coefficient, feed rate, and the angle of the mandrel slope. Using ANOVA, they evaluated the contribution of each parameter to the spinning process, thickness variation, and wrinkle severity. The findings revealed that the mandrel slope, support roller width, and spinning depth significantly influence the spinning process. Furthermore, the mandrel slope, roller radius, and spinning depth were recognized as critical factors affecting thickness variation. Regarding wrinkle severity, the roller radius, support roller width, spinning radius, and spinning depth were determined to have a considerable impact [50].

Dar et al., (2023) utilizes the Modigliani and Miller model alongside the Taguchi method to examine the effects of cost of capital, dividends, and initial share price on the share price at the end of a period. Employing Taguchi's L9 orthogonal array DOE, the study seeks to identify the most influential factor affecting the share price. ANOVA and analysis of mean are performed to ascertain the optimal combination of input factors and their respective contributions to the share price, with all analyses executed using MINITAB 18 software [51].

In this study, the authors will examine various domains where Taguchi orthogonal arrays have been utilized, including healthcare, engineering, physics, chemistry, finance, and economics. Furthermore, the study will pinpoint potential areas for future application of the Taguchi method across diverse fields.

5 Taguchi Method in Different Fields

Taguchi is a statistical method, which is used to study the examination of various parameters at a time on the response variable at different levels with a minimum number of trials. Based on given input parameters, we can easily predict the output value and develop the mathematical models through experimental trials. This paper aims to study the various fields where Taguchi was used. Taguchi gives an exact and successful procedure for design optimization. It is usually used to improve the process or product. Taguchi gives an arrangement similar to orthogonal arrays that give particular optimal settings of parameters and their levels for each examination. Note the number of trials conducted using the Taguchi method is very minimal. Taguchi orthogonal array methods are used in various fields such as Engineering, physics, chemistry, finance and economics, energy science etc. The authors are going to discuss some fields where Taguchi was used.

5.1 Engineering

Liu et al. (2019) investigated the effects of four factors at three levels on green roofs. The factors that impact green roofs are substrate depth, slope, substrate material and vegetation. The Taguchi L9 orthogonal array was used for the experimental setup and the S/N ratio and ANOVA were used to identify which factor impacts more on the green roofs. It was found that substrate material impacts more on green roofs. The percentage contribution of substrate depth, slope, substrate material and vegetation is 26.2%, 15.1%, 53.9% and 4.8% respectively on green roofs [52].

Athreya and Venkatesh (2012) investigated the effects of feed rate, depth of cut and cutting speed on lathe-facing operations at different levels. The Taguchi L9 orthogonal array method, S/N ratio and ANOVA were used to analyze the result. It was found that the cutting speed is more significant on the response variable. The contribution of feed rate, depth of cut and cutting speed is 15.65%, 18.56% and 58.49% respectively at different levels of lathe-facing operation. It means that the speed contribution is more significant on the response variable than the other two factors. Also, it was examined that the full factorial design and the Taguchi methods provided the same result [16].

Sadeghifam et al., (2019) investigated the impact of five controlled factors on energy optimization (EO). The factors are wall, floor, roof, ceiling and window. The Taguchi orthogonal and response means were used to analyze the result. It was found that the ceiling factor impacts more on EO. The ranking for the factors on EO is: rank 1 for the ceiling, rank 2 for the window, rank 3 for the wall, rank 4 for the roof and rank 5 for the floor [53].

Rodrigues et al. (2012) used Taguchi's approach to measure the effects of feed cut, speed and the depth of the cut on roughness and cutting force in turning mild steel using a high-speed steel cutting tool. Taguchi orthogonal array measures that the cutting speed had a significant effect on surface roughness and the feed rate and cut rate had a significant effect on roundness error. The factors that affect metal removal rate (MMR) are voltage, electrolyte concentration and feed rate. The feed rate affects more on MMR by approximately 59% by using the Taguchi L9 orthogonal array [54].

Munprom & Limtasiri (2019) investigated the effect of exposure resolution, exposure time per layer, layer height, post-cure time, up-speed, lift and post-cure temperature on flexural strength and hardness. Taguchi L8 was used for the experimental setup and a mean plot was drawn to identify which factor impacts more flexural strength and hardness. It was found that post-cure temperature impacts more on flexural strength and hardness [55].

Praveen et al., (2024) explores the optimization of CNC turning parameters for Cu-Al-Mn shape memory alloys (SMAs) produced via induction melting. The research aims to assess the impacts of feed rate, depth of cut, and cutting speed on surface roughness (Ra) and material removal rate (MRR). Employing Taguchi's DOE with an L9 orthogonal array, the study performs a performance analysis to pinpoint the optimal machining parameters. ANOVA is utilized to determine the primary factors affecting MRR and Ra. The findings underscore the effectiveness of the Taguchi method in optimizing CNC turning for Cu-Al-Mn SMAs, using tungsten carbide tip tools [56].

5.2 Physics

Karaagac and Köçkar (2019) examined the effect of different factors (temperature (degree Celsius)), reaction time (min), and hydrogen flow rate (ml/min)) on saturation magnetization. The Taguchi L9 orthogonal array method and S/N ratio were used to identify which factor impacted more on the saturation magnetization. It was found that the temperature (degree Celsius) had the most significant impact on the saturation magnetization. The ANOVA was used to measure the percentage contribution of each factor on saturation magnetization. The percentage contribution of temperature (degree Celsius)), reaction time (min), and hydrogen flow rate (ml/min are 97.98%, 1.25% and 0.48% respectively [57].

Mohamed and Lenin (2019) investigated the impact of factors (pulse off, current and pulse on) of the surface roughness on AA6082-T6. The experimental setup was developed with the Taguchi design. Taguchi's L9 orthogonal array method, average response and S/N ratio were used for the analysis. It was found that the "pulse on" had the most significant factor on the response variable than pulse off and current [58].

Vempati et al. (2018) examined the effect of various factors such as welding speed, welding current and welding voltage on improve welding strength. The Taguchi L9 method was used for the design of the experiment purpose. The S/N ratio and response mean were used to identify which factor impacts more on the welding strength. It was found that the welding voltage had a more significant impact on the welding strength and welding speed impacts very less on it [59].

Sobhani et al., (2018) investigated the impact of extinction coefficient, fin height, scattering albedo, rayleigh number, and fin length on the total Nusselt number on the cold wall. Taguchi L27 was used to develop the experimental setup. S/N ratio and ANOVA were used to identify which factor was more significant on the total Nusselt number on the cold wall. It was found that the Rayleigh number had the most significant on the total Nusselt number on the cold wall and fin height had impacts less on it [60].

Reddy et al. (2018) examined the impacts of Axial feed rate, pressure and friction coefficient on the bulge height and thinning distribution. The Taguchi L9 method, S/N ratio and ANOVA were used for this study. It was found that the pressure had the most significant impact on both bulge height and thinning distribution [61].

Park et al. (2017) examined the effect of various factors on powder-binder separation. The factors are initial powder volume, shear rate, powder size and injection temperature. The Taguchi orthogonal array and S/N ratio were used to identify which factor has the most significant impact on the powder-binder separation. It was found that the initial powder volume had the most significant factor on the above-mentioned response variable [42].

Celik et al. (2018) examined the impact of width, thickness, Reynolds number and pitch on friction. The Taguchi and ANOVA were used to measure the percentage contribution of each factor on friction. It was found that the percentage contribution of width, thickness, Reynolds number and pitch are 16.64%, 16.64%, 53.85% and 27.27% respectively [63].

5.3 Chemistry

Ahmad et al. (2012) investigated the effects of different factors on the mechanical properties of epoxy/nanoclay/multi-walled carbon nanotube (MWNT) nanocomposites. The factors are MWNT content, cure temperature, pre-cure temperature and nano clay content. The Taguchi orthogonal array method at three levels, Analysis of mean and Analysis of variance was used for the experimental setup and analysis. It was found that the pre-cure temperature, cure temperature and MWNT content had the most significant impact on the response variable [64].

Harlapur et al. (2018) examined the impact of sliding speed and load on the wear rate. The Taguchi method, S/N ratio, analysis of mean and analysis of variance were used to identify which factor impacts more on the response variable. It was found that the load factor had a major impact on the As-cast volumetric wear rate and 1 h volumetric wear rate than sliding speed [65].

Pouretedal et al. (2018) investigated the impacts of different factors on the recrystallization process of RDX particles and the recrystallization process of HMX particles. The factors are a ratio of solute to solvent/(g/ml), cooling ratio/ (degree celsius/min), a ratio of anti-solvent to solvent/ (V/V), and ageing time/h of the mixture. Taguchi L16 orthogonal array method was used at four levels for experimental setup, S/N ratio and ANOVA were used for the analysis. It was found that the ratio of anti-solvent to solvent/ (V/V) had the most significant on both the recrystallization process of RDX particles and the recrystallization process of HMX particles [66].

Madhavi et al. (2017) investigated the impact of factors on the optimization of selected properties like hardness and surface roughness. The Taguchi orthogonal array method, S/N ratio and analysis of variance were used to find which factor impacts more on the response variable. It was found that the material had a significant impact on the above response variable and other two factors such as cutting speed and feed rate had less impact on hardness and surface roughness [44].

Efficiency relies on maximizing emissions and engine performance for diesel engines with minimal experimentation. Dey et al., (2024) employs a fuzzy-assisted Grey Taguchi approach to examine the effects of engine load and fuel blend type on BSFC, NOx, UHC, and CO emissions. Fuzzy inference was utilized to consolidate multiple responses into a single objective, leading to the identification of optimal conditions with the D85B10E5 fuel blend at 100% load. ANOVA analysis revealed that engine load is the most significant factor. The findings suggest that, compared to traditional methods, the Grey-Fuzzy-Taguchi strategy is more effective in enhancing engine performance and reducing emissions [67].

5.4 Economics and Finance

Dar and Qadir (2019) investigated the effects of the value of the debt, the value of the firm, interest rate and volatility at one-time period on the probability of default and the default distance. The Taguchi L27, analysis of mean and ANOVA were used to identify which factor impacts more on the probability of default and the default distance. It was found that the volatility of assets impacts more on both the probability of default and the default distance. It was also found that there is a negative relationship between the probability of default and the default distance. The analysis of the mean identified the best optimal combination among the factors at different levels. It was found that when the value of a firm is high, the value of debt is low, volatility is high, and the interest rate is high at the constant time – it means this combination will provide a minimum probability of default and maximum distance to default [68].

Dar et al. (2018) investigated the effects of the value of the debt, value of the firm, interest rate and volatility at different periods on the probability of default. The Taguchi L27, analysis of mean and ANOVA were used to identify which factor impacts more on the probability of default. It was again found that the volatility of assets impacts more on the probability of default. The ranking of factors is rank 1: volatility, rank 2: the value of debt, rank 3: the value of a firm, rank 4: period and rank 5 interest rate. It was found that when the value of a firm is high, the value of debt is low, volatility is high, and the interest rate is high at the maximum time – it means this combination will provide a minimum probability of default. The optimal combination of those factors remains the same but if the data set changes then the ranking of those factors may be changed [69].

Dar & Anuradha (2018) examined the effects of factors (the underlying asset price, strike price, volatility, and interest rate) at a constant period on call option value by using the Black Scholes option pricing model. The Taguchi L9 orthogonal array method was used for the experimental setup and some statistical tests such as analysis of mean, S/N ratio, Tukey method and ANOVA were used to identify which factor impacts more significantly on the call option value. It was found that the underlying asset price impacts more on the call option value but it may be changed if the data set changes. The best combination among the factors at different levels is a higher underlying asset price, lower strike price, higher volatility and a higher interest rate at constant time [70].

Wang and Huang (2007) used the Taguchi method to calibrate the controllable parameters of a forecasting model. It was found that the Taguchi method with S/N ratio provides the ranking for the controllable parameters and it allows decision-makers to achieve the goal [46].

Dar & Anuradha (2018) examined the effects of factors (the underlying asset price, strike price, volatility, and interest rate) at different periods on put option value by using the Black Scholes option pricing model. The Taguchi L27 orthogonal array method was used for the experimental setup and some statistical tests such as analysis of mean and ANOVA were used to identify which factor impacts more significantly on the put option value. It was found that the underlying asset price impacts more on the call option value but it may be changed if the data set changes. The best combination among the factors at different levels is a higher underlying asset price, lower strike price, higher volatility, higher interest rate and higher period. Dar & Anuradha (2020) examined the factors that impact call option value at a constant interest rate. This Binomial option pricing model was used to estimate the value of the call option. Taguchi method, analysis of mean and ANOVA were used to identify which factor impacts more on the European call option value [71].

Mohammed et al., (2024) examines the valuation of Goodwill (GW), a crucial issue in financial analysis, particularly in the context of mergers and acquisitions where intangible assets are significant. By utilizing the Capitalization Method of Super Profit (CMSP), this research improves GW valuation through the incorporation of statistical tools and Taguchi method. The study seeks to identify key variables such as average profit, capital employed, and rate of return that notably affect GW valuation. Through regression analysis, correlation matrices, response analysis, and ANOVA, the research offers a comprehensive understanding of both internal and external factors influencing GW, providing valuable insights for informed decision-making [72].

5.5 General Science

Ketkar & Vaidya (2014) examined the effect of factors such as class 12, work experience, class 10, scores at qualifying degree, group exercise score, entrance test marks, and personal interview scores on MBA admission. Taguchi L8 method and S/N ratio were used for this purpose. It was found that group exercise, personal interviews and tests have positive gains whereas grades have a marginally positive gain also it was found that the 10th, and 12th marks and experience have a negative impact on admission [44].

Zahraee et al. (2015) investigated the effect of factors the arrival rate of donors, maximum inventory level, minimum inventory level and blood delivery policy on the blood supply chain efficiency. The Taguchi method was used to improve the blood supply chain efficiency. It was found that all factors are located at a high level except for the factor minimum inventory level [73].

Li and Zhu (2019a) adopted a hybrid multicriteria decision-making (MCDM) technique that merged the analytic hierarchy process (AHP) and grey relational analysis (GRA) to convert the indices of multiple UX quality characteristics into an MPCI and then used the Taguchi method to optimize UX [74].

Li and Zhu (2019b) used the fuzzy analytic network process-based Taguchi orthogonal array method to optimize the UX of cell phone app design. It was found that Home navigation was the most important factor for the multi-performance characteristic index [75].

Zhang and Wang (2019) utilized the Taguchi technique (L8) for streamlining the framework to upgrade the energy harvesting ability, which includes expanding the peak power output ratio of the framework and broadening the energy harvesting bandwidth. Given the consequences of the Taguchi technique, the impacts of every parameter on the peak power output ratio and energy harvesting bandwidth have been measured and thought about. It is discovered that in the frequency range of 1–30 Hz, the improvement of three parameters can double the energy harvesting bandwidth and increase the peak power output ratio [76].

Keleştemur & Arici, (2019) examined the effects of different factors on wear loss, compressive strength and ultrasonic pulse velocity (UPV). The factors are coarse scale, cement dosage, fine-scale and water/cement level that impact the above-mentioned response variables. The Taguchi orthogonal array method and the ANOVA were used for the experimental setup and data analysis respectively. It was found that when the cement dose increased, the water/cement ratio decreased that time compressive strength sample increased. The relationship between the compressive strength, wear resistance and UPV of mortar samples developed by substituting river sand with steel waste was investigated. It was found that the contribution of 13.95% of fine-scale, 27.38% of cement dosage, 14.78% of coarse-scale and 35.3% of water/cement ratio on wear resistance of cement mortar specimens [77].

6 Limitations

Taguchi's Orthogonal Array (OA)-based experimentation methodology has been widely used in industries for its simplicity and efficiency in conducting designed experiments. However, it's important to acknowledge its limitations to provide a balanced perspective on its applicability.

1. Limited to Orthogonal Arrays: Taguchi Methods rely heavily on the use of orthogonal arrays, which have a predetermined number of levels for each factor and may not always align with the actual factors under investigation. This limitation can restrict the flexibility of experimental designs and may not fully capture the complexity of real-world systems [11] (Roy, 2001).
2. Focus on Robustness: While Taguchi Methods excel in designing experiments that are robust to noise factors, they may not be as effective in uncovering the main effects and interactions between factors. This can be problematic when the goal is to understand the underlying relationships between variables rather than just achieving robustness [78] (Montgomery, 2017).
3. Assumption of Linearity: Taguchi Methods often assume linear relationships between factors and responses, which may not hold in many practical situations where non-linear effects are present. This can lead to inaccurate modelling and suboptimal process optimization [11] (Roy, 2001).
4. Lack of Statistical Rigor: Compared to more traditional statistical DOE techniques, Taguchi Methods may lack the same level of statistical rigour in terms of hypothesis testing and model validation. This can lead to uncertainty in the conclusions drawn from the experiments [79] (Myers et al., 2016).
5. Inadequate for Complex Designs: In scenarios where the factors are numerous and highly interrelated, Taguchi Methods may not be capable of capturing the full complexity of the experimental space. This can result in oversimplified models that fail to accurately represent the system under study [78] (Montgomery, 2017).

While Taguchi Methods offer advantages in terms of simplicity and efficiency, especially in industrial settings, researchers and practitioners should be aware of these limitations and carefully consider the appropriateness of this approach for their specific objectives and experimental conditions.

7 Conclusions

The Taguchi method is a very popular method for accurate results with fewer errors. It is an alternative method of FFD and it reduces our time, resources, number of trials etc. The procedure portrays the connection between the info and the yield item/process. It is utilized for improving frameworks (the framework might be straightforward or complex). It tends to be utilized to grow new items and used to improve the current items/forms. DOE is considered as the best strategy for the specialists; it directs a trial monetarily, gathers the data lastly discover essential targets and make an end from the examination. Its motivation is to accumulate information that can be analyzed by the statistical tools. It helps to improve an existing product/process or develop a new product. Mostly the Taguchi method was used in every field that we already discussed above. But there are other fields where Taguchi method was used such as advertise and marketing, biotechnology, health care and so forth. And also there are some fields where we can use the Taguchi method to improve a product/process. The Taguchi can be used in other fields such as Afzal & Kavitha (2018) mentioned that the impacts of different factors such as throughout, degree of balance, number of migration and resource utilization on the QoS will be studied in future [80]. Das & Goswami (2019) mentioned that the Taguchi method can be used to identify the effect of different factors on entrepreneurial networks on small firm performance in future [81]. Saidu (2019) also mentioned in his study that we can use the Taguchi method in order to find the effect of different factors on CEO characteristics and firm performance: focus on origin, education and ownership [82]. So, in nutshell, we can say that the Taguchi orthogonal method can be used in every field with less number of trials and it provides an accurate result with less error.

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Declarations

Competing Interests The authors declare no competing interests.

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